Parpadeador LED NE555

Halo

2026-09-20

1. Resumen

Este informe recoge el estado del proyecto tal como lo registra el Halo.

2. Esquemático

El esquemático `parpadeo555.kicad_sch` tiene 9 componentes, ['uuid': 'aaacfc79-7877-48d6-8dc9-7f418e82d9e7', 'pts': [[43.18, 76.2], [55.88, 76.2]], 'uuid': '52f0477a-d7f8-45c1-9727-859ffcd58e48', 'pts': [[55.88, 76.2], [55.88, 73.66]], 'uuid': '9b511a9a-ef63-40a9-8eab-420c5aa74692', 'pts': [[48.26, 76.2], [48.26, 73.66]], 'uuid': '2a8a6ba2-fb0d-43fb-8f66-673bd194e427', 'pts': [[43.18, 78.74], [55.88, 78.74]], 'uuid': '8e82586c-670d-42e9-a7dd-3f0143d0a8e8', 'pts': [[55.88, 78.74], [55.88, 81.28]], 'uuid': 'c041a596-02ec-4fd3-b85c-b5b9d2f83d97', 'pts': [[48.26, 78.74], [48.26, 81.28]], 'uuid': '3c4b0cad-7232-4327-a91d-df8066cdf0bf', 'pts': [[114.3, 80.01], [114.3, 85.09]], 'uuid': 'ce5714c8-47a1-4cd2-bd95-c72896961872', 'pts': [[129.54, 83.82], [114.3, 83.82]], 'uuid': '2eeab108-25de-46ee-a5f3-cdbb154f672a', 'pts': [[114.3, 92.71], [114.3, 97.79]], 'uuid': 'e55b073c-f6ee-47cc-a52f-15eac5ea85e9', 'pts': [[129.54, 88.9], [124.46, 88.9]], 'uuid': '9ae41db2-61cd-409a-ad70-839ecb6bc902', 'pts': [[124.46, 88.9], [124.46, 95.25]], 'uuid': 'a18cf12b-4024-4e50-a041-4a7843619f46', 'pts': [[124.46, 95.25], [114.3, 95.25]], 'uuid': 'f682c6f9-f4c5-4369-ac5e-d96eec565b68', 'pts': [[129.54, 91.44], [124.46, 91.44]], 'uuid': 'bac192f0-af2b-4afe-b887-f4a40ff5926a', 'pts': [[139.7, 76.2], [139.7, 66.04]], 'uuid': '2a3059f7-48c4-400e-aac0-bb390646a289', 'pts': [[139.7, 66.04], [167.64, 66.04]], 'uuid': 'ebf15d62-903f-456a-bd1d-6b07341e510e', 'pts': [[167.64, 66.04], [167.64, 77.47]], 'uuid': 'c9204535-0bb6-4356-ac74-3cce47ec89fc', 'pts': [[114.3, 72.39], [114.3, 69.85]], 'uuid': '41d98cac-190e-4647-a75b-c63bc66b2e7b', 'pts': [[129.54, 81.28], [127.0, 81.28]], 'uuid': 'b4247719-08eb-40f5-8db2-42c56e778fc3', 'pts': [[127.0, 81.28], [127.0, 78.74]], 'uuid': '3f1e6f69-5e99-4d15-9465-eb30cff63fb7', 'pts': [[142.24, 76.2], [142.24, 73.66]], 'uuid': '4be7074c-9654-417b-bb9d-896fd8eac2ab', 'pts': [[139.7, 96.52], [139.7, 99.06]], 'uuid': '1ec283ea-5c01-4a1a-a1c7-aa1ca3678aef', 'pts': [[114.3, 105.41], [114.3, 107.95]], 'uuid': '82c584e1-2821-4bc9-906e-28b4759c49b4', 'pts': [[167.64, 85.09], [167.64, 87.63]], 'uuid': '05c5ab96-f4cf-431c-9176-1eaaab900efa', 'pts': [[177.8, 77.47], [177.8, 74.93]], 'uuid': '6e162503-f657-41ac-b85c-1dd48acbdaf1', 'pts': [[177.8, 85.09], [177.8, 87.63]], 'uuid': '54734e84-25cd-4da8-adc9-9f0fc3b31758', 'pts': [[149.86, 86.36], [152.4, 86.36]], 'uuid': '3f1436bf-5a56-438f-9c11-52850ae35eb9', 'pts': [[219.71, 86.36], [214.63, 86.36]], 'uuid': '63354c4e-8727-4c3f-ba94-0a0829e97a1c', 'pts': [[227.33, 86.36], [236.22, 86.36]], 'uuid': '15ad24c9-685f-4728-837e-6c516e0a54a8', 'pts': [[236.22, 86.36], [236.22, 91.44]], 'uuid': '6f4c92a5-c267-4d5d-8641-005419811e17', 'pts': [[236.22, 99.06], [236.22, 101.6]]] cables y 3 bloques. Verificación de KiCad: ERC: 0 errores, 0 avisos; netlist idéntica a la intención; dibujo: 0 problemas.

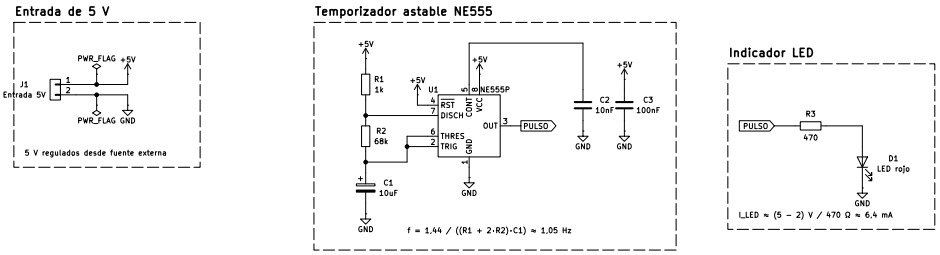


Figura 1: Esquemático de Parpadeador LED NE555.

2.1. Bloques del circuito

- **Entrada de 5 V** (J1)
- **Temporizador astable NE555** (C1, C2, C3, R1, R2, U1)
- **Indicador LED** (D1, R3)

2.2. Componentes

Referencia	Valor	Símbolo
C1	10uF	`Device:C_Polarized`
C2	10nF	`Device:C`
C3	100nF	`Device:C`
D1	LED rojo	`Device:LED`
J1	Entrada 5V	`Connector_Generic:Conn_01x02`
R1	1k	`Device:R`
R2	68k	`Device:R`
R3	470	`Device:R`
U1	NE555P	`Timer:NE555P`

3. Simulación

Simulación	Análisis	Estado	Criterios
555 · temporización astable ideal	tran	completado	cumple

3.1. Medidas

Simulación	Medida	Valor	Estado
555 · temporización astable ideal	frecuencia	1.05263 Hz	cumple

4. Placa de circuito impreso

La placa `parpadeo555.kicad_pcb` mide 41.5 × 27.5 mm, tiene ['F.Cu', 'B.Cu'] capas y está en etapa *ruteada*. Lleva 9 huellas (9 arriba, 0 abajo), 50 tramos de pista y 0 vías. Verificación: DRC: 0 errores, 0 avisos; 0 conexiones pendientes; paridad con el esquemático exacta.

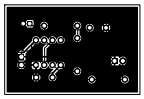


Figura 2: Cobre, contorno y serigrafía de la placa.

Zonas de cobre: GND en B.Cu.

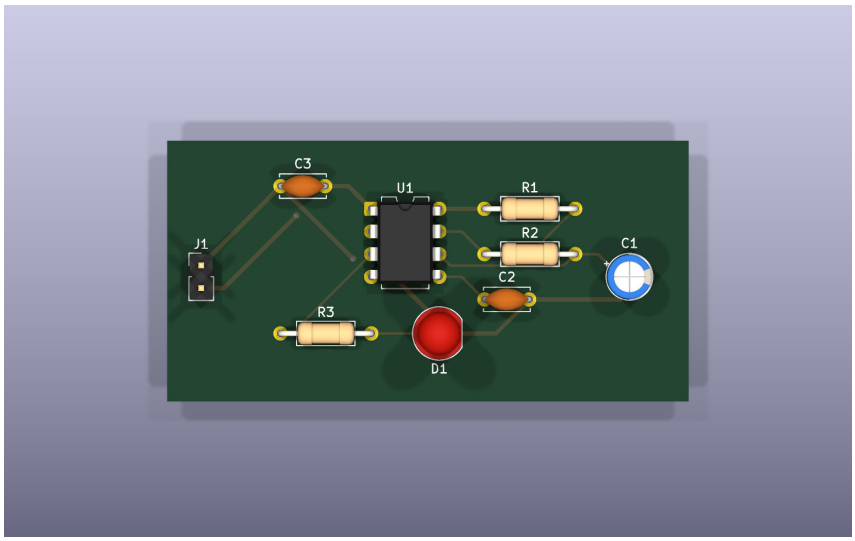


Figura 3: Vista 3D (arriba).

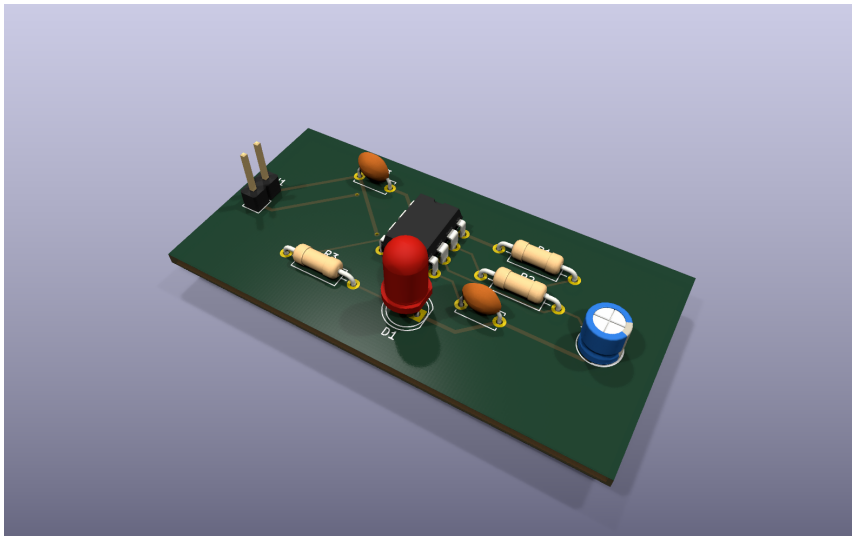


Figura 4: Vista 3D (perspectiva).

5. Fabricación

Todavía no se han preparado archivos de fabricación.

6. Conclusiones

Pendiente.